

Darsh Butala

Mumbai, India | darshbutala.dc@gmail.com

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Aerospace Engineering postgraduate at IIT Bombay with expertise in high-performance propulsion system design and control system development. Experienced in experimental combustion (DDT & Hybrid Rockets), CFD-based optimization, dynamic modelling, and flight stability analysis using MATLAB/Simulink. Seeking opportunities across space, defence, aero-engine, or UAV applications involving propulsion and flight dynamics.

SKILLS

Simulation & Engineering Analysis — ANSYS Fluent, ANSYS Mechanical, MATLAB & Simulink

CAD & System Integration — SolidWorks, CATIA, AutoCAD

Programming — Python, Java, C / C++

INTERNSHIPS & PROJECTS

IIT Bombay

Mumbai, India

M. Tech Thesis: LPG-Fuelled Deflagration-to-Detonation Transition (DDT) Jul 2025 – May 2026

- Designed and fabricated a compact LPG-air DDT test section with controlled ignition and purge cycles ensuring safe and repeatable operation.
- Conducted combustion experiments capturing high-frequency pressure-time data to characterize detonation wave structure and transition behaviour.
- Analysed shock dynamics and acoustic impulse generation for propulsion-based field applications.
- Integrated combustion theory, gas dynamics, and experimental diagnostics for system-level validation.

IIT Bombay

Mumbai, India

M. Tech Research Work: Hybrid Rocket Regression Study Jan 2025 – May 2025

- Conducted comparative analysis of GOX and N₂O oxidizers for HTPB-based hybrid propulsion systems.
- Evaluated regression rate limitations and investigated nano-aluminium additives for ~25% potential regression enhancement.
- Assessed performance trade-offs in hybrid motor design through literature synthesis and analytical modelling.

Thrust Vector Control (TVC) Model Rocket

Jan 2024 – May 2024

B. Tech Project

- Designed and simulated a servo-actuated thrust vector control mechanism for trajectory correction.
- Performed stability and flight dynamics analysis using MATLAB and Open Rocket.
- Fabricated and validated a working prototype demonstrating controlled thrust vectoring.

Design & Analysis of 100 kN LOX–Methane Rocket Propulsion System

- Performed chemical equilibrium and performance analysis using NASA CEA, estimating chamber conditions, thrust, and specific impulse.
- Generated optimized nozzle contours using Rocket Propulsion Analysis (RPA) for targeted expansion ratios and operating conditions.
- Conducted CFD simulations in ANSYS Fluent to evaluate flow expansion and performance efficiency.
- Designed cryogenic propellant tanks in SolidWorks and performed structural validation in ANSYS Mechanical under 12 atm internal pressure.
- Integrated engine, tank, and structural subsystems into a modular propulsion stage architecture.

Design and Control of Quadcopter Drone

- Developed mathematical models of quadcopter dynamics in MATLAB/Simulink and implemented a PID-based control architecture for stabilization.
- Designed a lightweight frame in SolidWorks, validating stability and response characteristics through simulation.

STAR Lab, Surat

Surat, India

Rocket Testing & Avionics Intern

Jun 2022 – Jul 2022

- **Static Test Pad Development:** Designed and implemented a static test platform for solid and hybrid rocket motors. Integrated load cell-based thrust measurement system for real-time performance monitoring. Established safety mechanisms including remote ignition, blast shielding, and emergency shutdown protocols.
- **High-Powered Rocket Avionics:** Developed onboard avionics integrating GPS, telemetry, and recovery systems. Programmed Arduino-based data logging for altitude, acceleration, and orientation tracking.

EDUCATION

Indian Institute of Technology Bombay (IIT B)

Mumbai, India

M. Tech, Aerospace Engineering (Specialization: Propulsion) | CPI: 8.23

2024 – 2026

Amity University Mumbai

Mumbai, India

B. Tech, Aerospace Engineering | CGPA: 8.78

2020 – 2024

Scholastic Achievement: Secured All India Rank 106 in GATE 2024 (Aerospace Engineering)

ADDITIONAL INFORMATION

Certifications: Simulation using ANSYS Fluent (Udemy), ANSYS Workbench (Udemy), Learning SOLIDWORKS (CAD/CIM Technologies), MATLAB & Simulink Onramp (MathWorks), Space Entrepreneurship 101 (Udemy), Introduction to Tech Entrepreneurship (Coursera)

Positions of Responsibility:

Teaching Assistant – Gas Dynamics, IIT Bombay

Jul 2025 – Dec 2025

- Guided 30+ students through assignments, evaluations, and technical doubt resolution.